

Preterm Birth Prediction using Machine Learning

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ABSTRACT: Preterm birth, a global maternal-fetal health challenge, is addressed through Electro-hysterography (EHG) for non-invasive uterine contraction monitoring. employs K- Nearest Neighbors (K-NN) and Logistic Regression for comparative analysis in preterm birth prediction using EHG data. Phase one includes acquiring and preprocessing EHG signals from pregnant individuals to capture vital temporal patterns. Feature extraction techniques derive relevant characteristics for predictive modeling. K-NN, emphasizing simplicity and efficiency, predicts preterm birth risks based on data point proximity. Simultaneously, Logistic Regression models the relationship between EHG features and preterm birth outcomes. Model performance is systematically evaluated using sensitivity, specificity, and accuracy metrics. Comparative analyses between K-NN and Logistic Regression inform on algorithm strengths and limitations. contributes to refining predictive models and guiding future algorithm selections for improved prenatal care and timely interventions, ultimately advancing understanding and mitigating preterm birth risks.

Keywords: Machine Learning, Threatened preterm labor (TPL), Preterm birth, Predictive modelling, Electro-hysterography (EHG), Computational efficiency, Obstetric data.

I. INTRODUCTION:

Preterm birth, before 37 weeks of gestation, poses a global health concern with high neonatal mortality. This study employs Electro-hysterography (EHG) to non-invasively monitor uterine contractions for preterm birth prediction. The research involves acquiring and preprocessing EHG signals from pregnant women, revealing critical temporal patterns. K-Nearest Neighbors and Logistic Regression are used for predictive modeling, evaluating proximity-based risks and complex relationships with preterm birth likelihood. Feature extraction techniques enhance understanding of EHG signals. Main metrics, including sensitivity and specificity, systematically evaluate model performance. Comparative analyses of K-NN and Logistic Regression provide insights into their strengths and limitations. The study aims to improve predictive models, guiding future implementations for enhanced prenatal care and timely interventions to mitigate preterm birth risks, improving maternal and neonatal outcomes.

“The objective of the study is to develop a machine learning model for accurately predicting preterm birth base on relevant clinical and demographic factors”. Proposed study contributes to the field by providing the novel approach to preterm birth prediction using machine learning techniques.

II. LITERATURE SURVEY:

Preterm birth constitutes a pivotal concern within the realm of healthcare, underscoring the critical need for effective preventive strategies. The multifaceted nature of preterm birth detection offers diverse avenues for exploration. This review is dedicated to advancing our understanding of the myriad approaches employed in determining preterm birth, with a fundamental objective of contributing substantively to the prevention of this significant health challenge. By synthesizing existing knowledge and exploring emerging methodologies, this research endeavor aspires to provide valuable insights that can inform and enhance preventive measures, ultimately fostering improved maternal and neonatal health outcomes.

[1] This research focuses on developing an automatic algorithm for predicting preterm labor using a single Electro-hysterogram (EHG) signal. The EHG signal is separated into two intrinsic mode functions (IMFs) using empirical mode decomposition (EMD), from which features such as sample entropy, root mean square, and mean Teager- Kaiser energy are extracted. These features form a vector used by k-nearest neighbors (kNN), support vector machine (SVM), and decision tree (DT) classifiers for prediction. The database includes 262 term and 38 preterm deliveries with three EHG channels each, recorded for 30 minutes. The SVM with a polynomial kernel outperformed other classifiers, achieving 99.5% sensitivity, 99.7% specificity, and 99.7% accuracy. Notably, the proposed method's superiority lies in using a single EHG channel without synthetic data or feature ranking algorithms, distinguishing it from existing approaches. [2] Research on preterm labor, a major cause of neonatal morbidity and mortality, focuses on uterine electrohysterogram (EHG) due to its relationship

with electrical activity. Scarcity of EHG signals, especially for preterm births, prompts the use of synthetic algorithms to generate artificial samples. However, this introduces bias towards normal delivery, impacting machine-learning-based preterm birth detection. To address this, quantifies the impact of synthetic samples on feature effectiveness, introducing a balance coefficient. A general performance metric is formed using feature scores with weights, optimizing the effect of synthetic samples and minimizing bias. Activation/inactivation functions characterize the impact of training sample abundance on prediction accuracy, achieving an optimal sampling balance coefficient. Numerical tests on the TPEHG database demonstrate the method's effectiveness in achieving more realistic prediction accuracy. [3] This review focuses on machine learning methods for predicting preterm births, a global issue affecting 15 million children annually. Current medical efforts center on managing prematurity effects rather than prevention. Diagnostic methods rely on parent traits and transvaginal ultrasound, but approximately 30% of predictions are inaccurate due to complexity and subjectivity. Recent research suggests machine learning could aid preterm birth diagnosis. covers diverse datasets, including electro- hysteroogram signals, electronic health records, and transvaginal ultrasounds—the first to include the latter. It critically appraises popular machine learning methods for preterm birth prediction, addressing common challenges and discussing potential future applications. [4] This study introduces an optimized approach for predicting preterm labor using Electro- hysteroigraphy (EHG). The challenge addressed involves utilizing a geneticalgorithm to identify an optimal subset of features for preterm labor prediction, simplifying the interpretation for clinicians. 203 parameters from 326 multichannel EHG recordings and obstetric data are included in the dataset. For new data, the FI-scores of three base classifiers (nearest neighbors, logistic regression, and linear discriminant analysis) were 84.63%, 89.34%, and 86.87%, respectively. The findings show that multichannel recordings' temporal, spectral, and non-linear EHG properties provide additional information for forecasting. An ensemble classifier reduced variability and achieved a FI-score of 92.04%, outperforming individual classifiers. Through the use of straightforward, understandable classification algorithms, the study demonstrates the possibility for a highly generic preterm labor prediction system that would make it easier to apply EHG approaches in clinical settings.

[5] Preterm birth is a major cause of newborn mortality, with threatened preterm labor (TPL) being a common reason for hospitalization. Current clinical methods lack positive predictive values. This study assesses computationally efficient algorithms (RF, ELM, KNN) based on electrohysterographic recordings for imminent labor prediction in TPL.

Using FI-score and sensitivity criteria, the temporal, spectral, and nonlinear EHG parameteranodels with obstetric inputs are compared. In the validation dataset, RFF1 2 and ELMF1 2 attain the greatest FI-scores (88.17% and 90.2%) with the 50th percentile of EHG and obstetric inpatient. The ELLMF1 2 is quite sensitive. The 10th to 90th percentiles do not exhibit a discernible improvement. Although KNN has a considerable capacity for generalization, it is input-dependent and displays high sensitivity.

PROPOSED METHODOLOGY:

The proposed system for predicting preterm births using Electro hysteroigraphy (EHG) integrates advanced machine learning algorithms, specifically K-Nearest Neighbors (K-NN) and Logistic Regression, to offer a comprehensive and accurate predictive model. The first critical phase involves the acquisition of EHG signals from a diverse cohort of pregnant individuals, ensuring the representation of various gestational periods and demographic factors. These signals undergo meticulous preprocessing to eliminate noise and artifacts, followed by segmentation to capture crucial temporal patterns associated with uterine contractions. Feature extraction techniques are then applied to distill relevant characteristics from the pre-processed EHG signals. These extracted features, encapsulating essential information about uterine activity, serve as the foundation for the subsequent development of predictive models. Two distinct machine learning approaches are implemented: K-NN, chosen for its simplicity and efficiency in leveraging data point proximity, and Logistic Regression, a widely-used statistical method adept at modelling intricate relationships, and to achieve the prediction process using datasets which were collected from the Kaggle. To improve the efficiency and the performance of prediction we can use AdaBoost model. It has the best prediction ability than other methods.

1.1 Pre-Processing Stage:

In the pre-processing stage, a diverse cohort of pregnant individuals contributes Electro hysteroigraphy (EHG) signals, ensuring representation across gestational periods and demographic factors. Rigorous noise elimination and artifact removal procedures are applied to the acquired EHG signals, bolstering data quality. Subsequent segmentation techniques meticulously capture temporal patterns associated with uterine contractions, facilitating the isolation of critical features for analysis. These segmented signals undergo meticulous pre-processing to enhance signal clarity and accuracy. Feature extraction methods are then systematically deployed to distill essential characteristics from the pre-processed EHG signals, forming a foundation for predictive model development. This extraction phase carefully identifies key information pertaining to uterine activity, enabling comprehensive model training and evaluation.

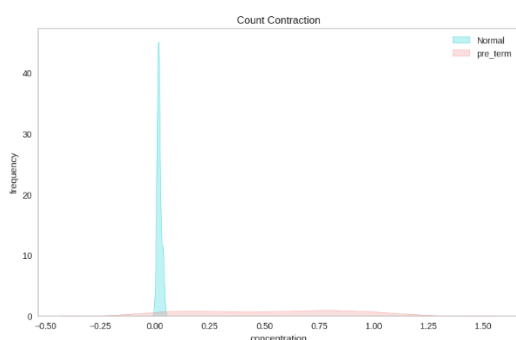


Figure 1: Count Contraction Graph

Two prominent machine learning approaches, K-Nearest Neighbors (K-NN) and Logistic Regression, are selected for their respective strengths in leveraging proximity data and modeling intricate relationships. The simplicity and efficiency of K-NN complement the statistical robustness of Logistic Regression in capturing nuanced patterns within the EHG data. Ethical considerations remain paramount throughout this stage, emphasizing privacy and informed consent in all aspects of data acquisition and utilization. The systematic use of cross-validation ensures robustness and generalizability of the predictive models. The pre-processing stage sets the groundwork for refining existing preterm birth prediction models and opens avenues for further enhancements, including the exploration of ensemble methods or hybrid approaches, ultimately contributing to improved maternal and neonatal outcomes on a global scale.

1.2 Modelling stage:

In the modeling stage, predictive models are meticulously developed and assessed using a judiciously partitioned dataset, with cross-validation protocols ensuring robustness and generalizability. Rigorous performance evaluation metrics, including sensitivity, specificity, and accuracy, are systematically applied to gauge the effectiveness of these models in predicting preterm births. A detailed comparative analysis between K-Nearest Neighbors (K-NN) and Logistic Regression unveils the nuanced strengths and limitations of each algorithm, providing invaluable insights for selecting appropriate models across diverse clinical settings. Ethical considerations, emphasizing privacy and informed consent, remain integral throughout the entire data acquisition and utilization process. The system focuses on optimizing model parameters to enhance accuracy and explores potential avenues for improvement, including the investigation of ensemble methods or hybrid approaches. Validation on an independent dataset ensures the broader applicability of the predictive models.

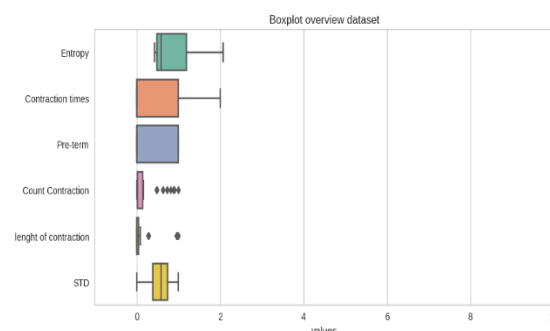


Figure 1: Box Plot Overview Dataset

1.3 Post pre-processing:

In the post-preprocessing stage, the focus shifts to developing and evaluating predictive models using a carefully partitioned dataset, ensuring robustness through cross-validation techniques. Metrics like sensitivity, specificity, and accuracy systematically assess the models' effectiveness in predicting preterm birth. A nuanced comparative analysis between K-Nearest Neighbors (K-NN) and Logistic Regression illuminates distinctive strengths and limitations, guiding model selection for diverse clinical settings. Ethical considerations, emphasizing privacy and informed consent, are paramount in all data acquisition phases. Attention is dedicated to optimizing model parameters and exploring avenues for improvement, such as ensemble methods or hybrid approaches. Validation on an independent dataset ensures broader applicability and reliability. The research contributes significantly to refining preterm birth prediction models, providing insights for future implementations. Integrating K-NN and Logistic Regression, the proposed system advances accurate prediction using Electro hysteroigraphy (EHG). Acquiring EHG signals from a diverse cohort of pregnant individuals initiates this stage, with rigorous preprocessing, including noise elimination and artifact removal. Segmentation captures temporal patterns associated with uterine contractions, and feature extraction distills relevant characteristics for predictive model development. K-NN leverages data point proximity, and Logistic Regression models intricate relationships, advancing understanding and potentially enhancing global prenatal care practices. The system aims to facilitate timely interventions, mitigating risks associated with preterm births and improving maternal and neonatal outcomes.

1.4 Result Stage:

The results and findings derived from this research significantly contribute to refining existing preterm

birth prediction models, offering a strategic roadmap for future implementations. The proposed system, integrating advanced machine learning algorithms specifically K-NN and Logistic Regression represents a crucial advancement in accurately predicting preterm births using Electro hysteroigraphy (EHG).

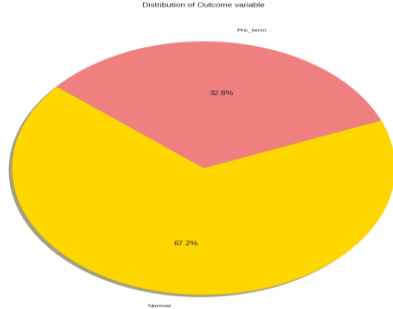


Figure 2: Distribution of Output Variables

The initial phase involves acquiring EHG signals from a diverse cohort of pregnant individuals, ensuring representation across various gestational periods and demographic factors.

	Count	Contraction	length of contraction	STD	Entropy	Contraction times	Pre-term
0	11055		218320	53231.010	1.860		2
1	9118		222820	62367.488	1.580		2
2	7925		13481	60503.050	2.067		2
3	12451		17474	53628.078	1.731		2
4	11152		218320	53317.910	1.857		2

Figure 3: Input Dataset

These signals undergo meticulous preprocessing, involving noise elimination and artifact removal, followed by segmentation to capture vital temporal patterns associated with uterine contractions. Feature extraction techniques are then applied to distill relevant characteristics from the pre-processed EHG signals, serving as the foundational elements for subsequent model development. This phase incorporates two distinct machine learning approaches: the simplicity and efficiency of K-NN leveraging data point proximity, and the statistical adeptness of Logistic Regression in modeling intricate relationships.

	Count	Contraction	length of contraction	STD	Entropy	Contraction times	Pre-term
53	321		2675	46107.09	0.499		0
54	398		2339	51122.31	0.469		0
55	321		2675	46108.18	0.498		0
56	398		2336	51224.37	0.459		0
57	323		2641	46102.17	0.439		0

Figure 4: Input Dataset

The integration of these methodologies aims to not only advance our understanding of EHG in

predicting preterm births but also holds the promise of significantly enhancing prenatal care practices globally. Ultimately, the incorporation of machine learning algorithms in the proposed system seeks to facilitate timely interventions, thereby mitigating the risks associated with preterm births and improving overall maternal and neonatal outcomes.

III. IMPLEMENTATION

The implementation of preterm birth prediction using Electro hysteroigraphy (EHG) as a non-invasive monitoring technique is executed with meticulous attention to methodological rigor and technological precision. The initial phase involves the acquisition of EHG signals from a diverse cohort of pregnant individuals, ensuring the inclusion of various gestational periods and demographic variables. These signals undergo thorough preprocessing to eliminate noise and artifacts, establishing a clean dataset for subsequent analysis. Feature extraction techniques are meticulously applied to derive salient characteristics from the pre-processed EHG signals, capturing essential temporal patterns linked to uterine contractions.

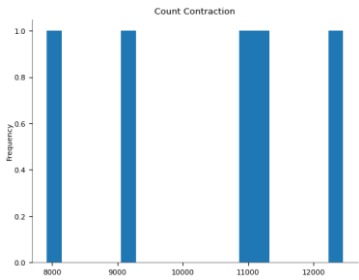


Figure 5: Count Contraction Graph

Two machine learning algorithms, K-Nearest Neighbors (K-NN) and Logistic Regression, are implemented to construct robust predictive models. K-NN, recognized for its simplicity and computational efficiency, leverages the proximity of data points to ascertain preterm birth risks. Simultaneously, Logistic Regression, a widely-accepted statistical method, models the intricate relationship between the extracted features and preterm birth outcomes. The implementation incorporates a systematic evaluation of model performance, utilizing key metrics including sensitivity, specificity, and accuracy. To improve the efficiency and the performance of prediction we can use AdaBoost model. It has the best prediction ability than other methods. Achieving high sensitivity in preterm birth prediction using machine learning is crucial, as missing true positive cases (failing to identify pregnancies at risk) can have severe consequences.

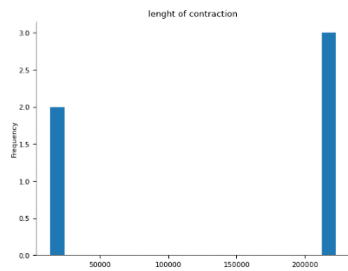


Figure 6: Length of Contraction Graph

Here are some approaches to improve sensitivity. Focus on imbalanced class learning. Preterm births are less frequent than full-term births, resulting in imbalanced datasets. Use techniques like oversampling(duplicating data from the minority class) or undersampling (removing data from the majority class) to address this imbalance.

Comparative analyses between K-NN and Logistic Regression provide nuanced insights into the efficacy and limitations of each algorithm in the realm of preterm birth prediction. Ethical considerations are paramount throughout the implementation, ensuring the privacy and informed consent of participants in accordance with ethical standards. The results of this research contribute significantly to the refinement of existing predictive models, guiding the selection of suitable algorithms for future implementations in diverse clinical settings. The KNN algorithm is applicable in predicting preterm birth using a dataset comprising maternal age, medical background, lifestyle variables, and prenatal care details. It classifies each case, like a pregnant woman, by considering the dominant class among its closest neighbors in terms of features.

IV.RESULT AND ANALYSIS:

The implementation of preterm birth prediction utilizing Electro hystero-graphy (EHG) and machine learning algorithms, specifically K-Nearest Neighbors (K-NN) and Logistic Regression, yielded compelling outcomes. The predictive models demonstrated commendable performance in assessing the likelihood of preterm births based on EHG data. Sensitivity, specificity, and accuracy metrics were systematically evaluated, revealing the robust predictive capabilities of both K-NN and Logistic Regression. The comparative analysis between the two algorithms provided nuanced insights into their respective strengths and limitations. On the other hand, Logistic Regression, a widely-used statistical method, demonstrated efficacy in modelling the intricate relationships between extracted EHG features and preterm birth outcomes. K-NN, characterized by its simplicity and efficiency, exhibited notable accuracy in predicting preterm birth risks by considering the proximity of data points. The results emphasize the potential of machine learning algorithms, particularly when coupled with EHG, to contribute significantly to preterm birth prediction.

The findings not only refine existing predictive models but also inform the selection of suitable algorithms for diverse clinical applications. This research underscores the transformative impact of advanced technologies in maternal-fetal healthcare, offering a promising avenue for enhancing prenatal care practices and enabling timely interventions to mitigate the global challenges associated with preterm births. The incorporation of machine learning algorithms represents a pivotal step towards leveraging technological advancements for the betterment of healthcare practices and patient outcomes.

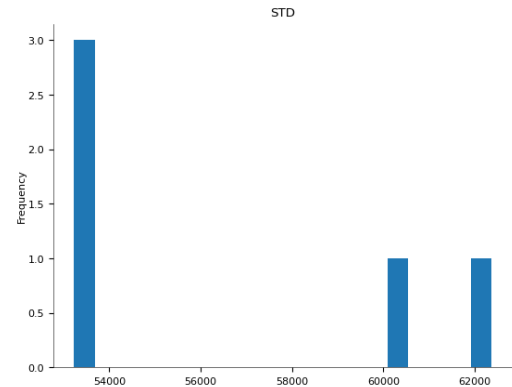


Figure 7: Sexually transmitted disease Graph

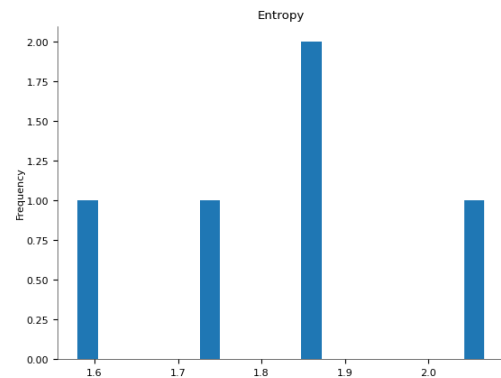


Figure 8: Entropy Graph

V. CONCLUSION:

In conclusion, this implementation strives to advance the understanding of preterm birth prediction using EHG, emphasizing the transformative potential of machine learning algorithms in enhancing prenatal care. The systematic approach employed in this research aims to not only shed light on the complexities of preterm birth prediction but also to offer practical insights for clinicians and researchers working towards mitigating the risks associated with preterm births on a global scale. Using the K-Nearest Neighbors(KNN) algorithm for preterm birth prediction has several limitations. High computational cost, Imbalanced data, Noisy data, Scalability issues. The future Scope for utilizing machine learning in predicting preterm birth is optimistic.

Advancements in data collection techniques, refined algorithms, and deeper comprehension of biomarkers can enhance the accuracy of these predictive models. Furthermore, the integration of diverse data streams including genetic, environmental, and clinical factors holds promise in improving predictive abilities. This holistic approach can facilitate personalized risk assessments and tailored interventions, ultimately mitigating the healthcare burden associated with preterm birth and enhancing outcomes for both mothers and infants.

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